

The Corporate Bond ETF Creation Puzzle

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Abstract

Standard ETF arbitrage dictates that authorized participants (APs) create shares only when the fund trades at a premium. I document a "creation puzzle" in corporate bond ETFs: across a sample of 211 U.S. funds, 15.7% of primary market creations occur while the ETF trades at a discount to its net asset value (NAV). Using daily fund holdings and bond transactions, I show that this phenomenon is driven by AP inventory constraints rather than traditional price arbitrage. Specifically, APs selectively deliver hard-to-price bonds subject to elevated inventory pressure into ETF creation baskets. Following delivery, these bonds underperform by an average of 5.5 basis points over the subsequent 20 trading days. Furthermore, relative to a matched sample of non-delivered bonds, the delivered assets experience significant dealer sell-offs. These findings suggest that APs strategically utilize the ETF primary market as a mechanism to offload burdensome inventory and avoid inventory risk, bypassing traditional over-the-counter (OTC) liquidations.

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1 Introduction

It has been increasingly suggested that financial intermediaries play an important role in providing valuable liquidity to investors. Indeed, the corporate bond exchange-traded fund (ETF) holding illiquid corporate bonds traded on the over-the-counter (OTC) market allows investors to trade ETF shares throughout the day easily on the stock exchange at a low cost.

Through the creation and redemption mechanism, ETFs maintain liquid secondary market trading while keeping prices closely aligned with their net asset values (NAV). The creation and redemption process relies on authorized participants (APs), which are typically large financial institutions. Specifically, APs create ETF shares by delivering a designated basket of underlying assets to the ETF issuer (creation basket), or redeem shares in exchange for the underlying portfolio (redemption basket). Unlike equity ETFs, where creation baskets and redemption baskets strictly mirror underlying portfolio weights, corporate bond ETF issuers allow custom baskets containing a subset of fund actual holdings to circumvent the high costs of trading the full bond portfolio. This discretion in basket composition, however, raises substantial agency problem concerns.

Under standard textbook arbitrage theory, ETF share creation is mechanically tied to secondary market premiums. When an ETF trades at a premium on the exchange to its net asset value (NAV), authorized participants (APs) can exploit this price discrepancy by purchasing the underlying baskets of securities in the OTC market or offloading from their own inventory to form the creation basket, delivering the basket to the ETF issuer with a fee to create new ETF shares, and selling those more liquid ETF shares at the higher market price on the exchange. This textbook arbitrage activity generates downward pressure on the ETF price and upward pressure on the underlying corporate bonds, thereby restoring parity. Consequently, existing literature and market wisdom widely treat share creations as an exclusive byproduct of ETF premiums, while redemptions are

presumed to resolve discounts. However, the empirical reality of corporate bond ETFs departs substantially from this stylized dichotomy.

Authorized participants (APs) link the two markets by delivering bond baskets for newly issued ETF shares or returning shares to redeem bonds. The usual arbitrage prediction is straightforward. If the ETF trades above its net asset value (NAV), the AP can buy bonds, create shares, and sell the shares at a premium. A discount suggests the reverse: buy ETF shares and redeem them for bonds. But APs also trade in the underlying bond market. For an AP that already owns bonds, creating shares is one way to dispose of those positions. Its decision depends on the cost of selling the bonds elsewhere as well as on the ETF premium.

I study share creations that occur after an ETF trades at a discount. In my sample of U.S. corporate bond ETFs, these transactions account for 15.66% of net creations with observed premiums. I find that the discounts close rapidly around creation. I also find that bonds inferred to enter the fund subsequently underperform similar bonds whose holdings remain unchanged. These results suggest that understanding discounted creations requires looking at the bonds exchanged, as well as at the price of the shares issued in return.

Corporate bond ETFs give APs some choice over which bonds to deliver. Creation baskets can contain a small subset of the fund's holdings, and the fund and its counterparties can negotiate their composition. This flexibility allows an AP to deliver bonds it already holds, which can make creation attractive even at a discount (Pan and Zeng, 2023). The fund, in turn, may want bonds that are costly to obtain in the dealer market. Du (2025) shows how existing dealer inventories can lower the cost of sourcing illiquid bonds. Both parties may therefore have reasons to agree to a discounted creation. How the delivered bonds subsequently perform is an empirical question.

I use daily holdings and fund information from ETF Global, ETF prices, and En-

hanced TRACE transactions to examine these exchanges. Increases in shares outstanding identify net creations. I infer bond deliveries from increases in holding quantities and compare each delivered bond with an unchanged holding in the same industry and the same ETF creation event. I match on issue size, maturity, coupon, rating, trading activity, and prior customer net selling. The final return sample contains 503,224 pairs across 4,292 creation events. Both bonds face the same fund and event conditions.

The average premium in the bond-event sample rises from -13.44 basis points on the trading day before creation to -0.86 on the recorded creation day and 0.65 on the following trading day. Both ETF prices and NAVs adjust: prices rise while NAVs fall. The same pattern appears when I require the same events to be observed throughout the window. The discount thus closes around positive net issuance, even though the initial discount would suggest redemption under the simple arbitrage account. The daily sequence does not establish that creation causes convergence, since trading and valuations can respond to the same news.

Delivered bonds earn lower subsequent clean-price returns than matched unchanged holdings. At return horizon 20, the mean difference is -1.28 basis points when returns are winsorized at common 0.5th and 99.5th percentile cutoffs. Adding bond and creation-event fixed effects, liquidity controls, and prior customer net selling gives an estimate of -1.36 basis points. The corresponding standard errors are 0.63 and 0.49 basis points. The estimates are also negative without winsorization, although the horizon-20 differences are no longer significant at the 5% level. Large return observations therefore affect the precision of the estimated underperformance.

The discount compares the ETF share price with the reported value of the portfolio; the return estimates compare bonds entering the fund with bonds already held. Prices can converge to NAV even if the new bonds are not representative of existing holdings. Nor is their subsequent underperformance a direct measure of AP profits or shareholder

losses. Measuring those gains and losses would require the terms of the exchange, the AP’s acquisition costs, and the value of the bonds to each party. Daily holdings can also include the fund’s own trades, and the Enhanced TRACE data used here do not identify individual dealers. The analysis therefore measures conditional return differences among inferred deliveries, without separately identifying inventory relief and informed bond selection.

This paper relates most closely to research on AP inventory and the choice of creation baskets. Pan and Zeng (2023) explain why inventory management can lead APs to trade against the direction suggested by the ETF premium. I document how often such creations occur and examine the accompanying changes in ETF valuations and bond returns. Reilly (2022) and Du (2025) study how the bonds selected for delivery affect the cost of liquidity transformation. I examine that selection when the conventional incentive to create at a premium is absent. The fund’s own decisions are part of the explanation: managers can use basket choices and direct bond trades to meet their portfolio objectives (Koont et al., 2025; Huang et al., 2026). A discounted creation can serve counterparties with different trading alternatives and portfolio needs.

Section 2 discusses related work. Section 3 describes the primary market, and Section 4 presents the data and empirical design. The results and robustness checks appear in Sections 5 and 6. Section 7 concludes.

2 Literature review

2.1 ETF arbitrage and liquidity transformation

Creations and redemptions allow trading demand in ETF shares to reach the underlying assets. Lettau and Madhavan (2018) describe this connection between the two markets. In equities, Ben-David et al. (2018) find that ETF ownership is associated with greater

volatility in the underlying stocks, and Brown et al. (2021) show that primary-market flows reveal non-fundamental demand and predict subsequent ETF returns. Primary-market activity thus contains information about the demand behind price movements, beyond the change in the quantity of assets held by the fund.

The cost of trading corporate bonds changes how ETFs accommodate that demand. Dannhauser (2017) finds higher valuations associated with ETF ownership but no general improvement in individual-bond liquidity. The distinction is especially relevant during market stress. Dannhauser and Hoseinzade (2022) document price pressure and subsequent reversals associated with fund flows during the Taper Tantrum. Shim and Todorov (2023) show how APs can absorb redeemed bonds into inventory to protect their existing positions and avoid fire sales. Whether ETF flows lead to immediate bond trading depends in part on intermediaries' willingness and capacity to hold the bonds.

Pan and Zeng (2023) make this inventory decision explicit. APs arbitrage ETF shares while managing their bond positions, and a sufficiently large positive inventory can make creation worthwhile at a discount. I examine the bonds transferred in these creations and their subsequent returns. My event-time evidence also complements their analysis of premium dynamics. Their regressions ask how inventory-related shocks predict future premiums; my plots trace average premiums around selected creations. Convergence in the latter can coexist with conditional persistence in the former.

2.2 Dealer inventory and bond selection

A dealer's cost of delivering a bond depends on whether it already holds the security and on its opportunities to trade elsewhere. In Duffie et al. (2005), search and bargaining affect both transaction timing and prices. Hendershott et al. (2020) show how repeat trading relationships affect execution and pricing in corporate bonds. An AP that must locate and purchase a bond can therefore face a different delivery cost from one that

holds the bond in inventory.

Holding inventory also exposes dealers to risk. Bessembinder et al. (2018) document lower capital commitment by bank-affiliated dealers alongside relatively small changes in average execution costs. Goldstein and Hotchkiss (2020) find that dealers often offset transactions quickly, particularly for riskier and less actively traded bonds. Dealers can thus adjust their trading and inventory in several ways. Customer selling may make bonds available for creation, but aggregate order flow alone does not reveal which AP holds them or why that AP chooses to deliver them.

Creation baskets give these inventory decisions a role in asset selection. Reilly (2022) finds that delivered bonds underperform and studies the cost to ETF investors. Du (2025) shows that dealers with inventory can source illiquid bonds more cheaply and reduce adverse selection. These explanations allow inventory-based exchange to benefit the fund even when delivery is selective. I compare delivered bonds with similar unchanged holdings during discounted creations. The comparison measures their relative performance; it leaves open whether a particular transfer reflects an inventory benefit or informed selection.

2.3 Portfolio demand by the ETF

The fund also chooses which bonds to exchange. Koont et al. (2025) show that managers of passive corporate bond ETFs actively select cash and bond baskets to balance index tracking with liquidity transformation. Huang et al. (2026) document direct bond trading by ETFs and show that it plays different roles in passive and active funds. Managers can therefore adjust portfolios through both the primary market and ordinary bond trades.

An ETF may accept a creation basket to obtain bonds it wants, while the AP accepts the same trade to dispose of bonds it owns. This gives the fund's demand a role in explaining discounted creations. It also affects measurement: an increase in reported

holdings can include a direct purchase as well as an in-kind delivery. I use holdings changes to infer additions around creation and examine their returns alongside ETF valuations. The daily data do not reveal each counterparty’s motive.

3 Institutional background

3.1 Trading shares and trading bonds

An investor can obtain corporate bond exposure by buying ETF shares from another investor. This secondary-market trade transfers existing shares without changing shares outstanding or requiring the fund to trade its bonds. In the primary market, an AP delivers a specified basket of bonds and possibly cash to create shares, or returns shares to redeem assets. ETF shares can consequently trade even when many of the underlying bonds do not (Lettau and Madhavan, 2018).

Arbitrage is profitable when the proceeds from selling one side of the exchange exceed the cost of acquiring the other. A sufficiently large premium can cover the cost of buying a basket and creating shares; a sufficiently large discount can cover the cost of buying shares and selling the redeemed assets. Fees, transaction costs, and executable bond prices determine whether the trade pays. NAV provides a reference value for the portfolio, but it does not guarantee that every bond can be traded at its recorded price.

3.2 Creation baskets

The fund initiates the standard basket process. The issuer or its agent prepares a portfolio composition file (PCF) that lists the securities and cash amounts in the basket. The agent sends this file to NSCC for distribution before the next trading day (DTCC, 2024). APs can then use the published basket to evaluate a creation or redemption.

The fund and the AP may agree on a custom basket that differs from the published

one. Corporate bond baskets often contain only a subset of the portfolio (Du, 2025), which avoids the need to trade every illiquid issue in a broad fund. The manager can request bonds that improve index tracking or meet other portfolio needs. The AP can propose bonds it already owns or can obtain cheaply (Koont et al., 2025). The exchange takes place only if the fund accepts the basket. The published PCF, the agreed basket, and the subsequent holdings change are therefore distinct records of the process.

3.3 Why create after a discount?

For an AP that holds bonds, creation is an alternative to selling them in the dealer market. A sale requires finding buyers and accepting their terms. Creation converts the bond position into ETF shares. Search costs and the value of reducing inventory can make that alternative attractive even when the shares trade below reported NAV (Duffie et al., 2005; Pan and Zeng, 2023). The ETF discount alone does not capture the AP's opportunity cost of delivering the basket.

The fund may benefit from obtaining bonds that would otherwise be costly to source (Du, 2025). It may also receive selectively delivered bonds with weaker subsequent returns (Reilly, 2022). Distinguishing these possibilities requires information about the bonds exchanged. A net increase in shares outstanding, by itself, cannot establish a benefit from intermediation or a loss to existing shareholders.

The daily data record only part of the transaction. Shares outstanding reflect net creation after any offsetting creations and redemptions. A bond holding can rise because of delivery, direct fund trading, or both (Huang et al., 2026). The discount before the recorded share increase can also differ from the intraday discount when the parties agreed to trade. The empirical definitions below follow what can be observed in these daily records.

4 Data and methodology

4.1 Data and sample

I combine daily holdings and fund information from ETF Global with ETF prices and corporate bond transactions from Enhanced TRACE (ETF Global, nd; FINRA, nd). The ETF panel covers January 4, 2016, through October 30, 2025. ETF prices come primarily from CRSP; when a CRSP price is missing, I use a screened price from the original fund panel (Center for Research in Security Prices, nd). Sample construction corrects discrete differences in the scale of prices and NAVs and screens changes in shares outstanding that resemble splits or short-lived reporting reversals. The final panel contains 255,044 ETF-date observations for 211 ETFs.

I match ETF records by standardized ticker and date, and bond records by CUSIP and observation date. Bond characteristics include offering amount, remaining maturity, coupon, credit rating, and industry. Enhanced TRACE records transaction prices and sizes, customer buy and sell direction, and whether trades involve dealers or customers. My data do not contain anonymous dealer identifiers. They support measures of aggregate customer order flow, but do not allow me to assign inventory or a delivery to a particular AP.

The full ETF panel measures the frequency of discounted creations. The bond analysis starts with 4,902 events classified as discounted in the source sample. In the cleaned ETF panel, 4,868 of these events have a negative baseline premium and 34 have a missing baseline premium. I retain all 4,902 for the bond and event-time analyses before imposing further requirements. Matching and return availability leave 4,292 events in the return sample. Table 1 reports these restrictions separately from the ETF frequency sample.

4.2 Identifying creations and deliveries

The ETF premium, in basis points, is

$$PD_{f,t} = 10,000 \left(\frac{P_{f,t}}{NAV_{f,t}} - 1 \right), \quad (1)$$

where $P_{f,t}$ is the share price and $NAV_{f,t}$ is NAV per share for ETF f on date t . Negative values indicate discounts. I identify a net creation when shares outstanding increase between consecutive dates on the project trading calendar. Day zero is the date of that recorded increase. The initial premium is measured on the preceding trading day, day -1 , so a discounted creation follows an observed discount. This definition does not recover the intraday terms of the AP's trade.

Within a creation event, I infer delivery when the reported quantity of a bond increases. Unchanged positions form the control pool; declining positions are excluded. Measuring quantities prevents a price change from being mistaken for a delivery. This follows the holdings-based approach in Reilly (2022) and Du (2025). Its accuracy depends on the reporting interval capturing the creation and on other portfolio trades not dominating the increase. Reporting delays, offsetting gross flows, and simultaneous fund trading can all affect the measure. Direct bond trading documented by Huang et al. (2026) is one reason to retain the distinction between inferred delivery and an observed AP transfer.

4.3 Matching within a creation event

I compare each delivered bond with an unchanged holding in the same ETF, creation event, and industry. This keeps the fund, event date, and surrounding market conditions common to both bonds. I choose the control using six pre-event characteristics: log offering amount, remaining maturity, coupon, average numerical rating, log turnover, and

average customer net selling over the preceding ten days. These characteristics describe bond structure, credit risk, trading activity, and recent order flow, all of which may be related to delivery and future returns.

The distance between delivered bond i and eligible control j is

$$d(i, j) = \left[\sum_{k=1}^6 \left(\frac{X_{ik} - X_{jk}}{s_k} \right)^2 \right]^{1/2}, \quad (2)$$

where X_{ik} denotes characteristic k and s_k is its pooled standard deviation before matching. I select the nearest control in the same event and industry, allowing a control to be used more than once. Each delivered bond contributes one pair, and all pairs receive equal weight. I do not impute missing covariates. Controls are chosen before I require future returns to be observed.

The initial match yields 599,235 pairs. Requiring finite returns for both bonds at all eight horizons leaves 503,224. The largest absolute standardized mean difference across the six characteristics is 0.076 using pre-match standard deviations and 0.081 using matched-sample standard deviations. Table 2 reports the means and standardized differences. Matching brings the groups close on these observables, although unobserved information and changing portfolio needs may still affect delivery. Appendix A gives further construction details and variable definitions.

4.4 Bond returns

I measure a bond's clean-price return relative to its price on the recorded creation date:

$$R_{b,e}(h) = 10,000 \left(\frac{p_{b,e}(h)}{p_{b,e}(0)} - 1 \right). \quad (3)$$

The subscripts b and e index bonds and creation events, and h denotes the return horizon. The price series uses the last trade of the day and carries the most recent price forward

when a price is missing. I retain the source horizons $h = 1, 2, 3, 4, 5, 10, 15, 20$. The underlying dates have not been reconstructed on a common calendar of fresh transactions, so these horizon labels are not verified trading-day counts. The returns exclude accrued interest and coupon payments, unlike the dirty-price returns used by Reilly (2022).

I first average the difference between delivered and control returns within pairs. To account for persistent bond characteristics and conditions at creation, I also estimate

$$R_{b,e}(h) = \alpha_b^h + \delta_e^h + \beta_h D_{b,e} + \gamma_h' Z_{b,e} + \varepsilon_{b,e}^h, \quad (4)$$

where $D_{b,e}$ equals one for an inferred delivery. Bond fixed effects α_b^h absorb persistent differences across securities, and creation-event fixed effects δ_e^h absorb conditions common to bonds in the same ETF creation. A negative β_h indicates that delivered bonds underperform conditional on these effects and the controls $Z_{b,e}$.

The first specification includes delivery status and the fixed effects. I then add pre-event log turnover and log Amihud illiquidity, a measure of absolute price changes relative to dollar trading volume (Amihud, 2002). The final specification also controls for customer net selling over the preceding ten days. This measures aggregate selling to dealers rather than the inventory of a particular AP. Controls enter in standard-deviation units. Reused bonds enter with their matching frequencies, preserving the weights in the paired comparison. The specifications with liquidity controls lose 74 pairs because Amihud illiquidity is missing or nonpositive.

I cluster standard errors by bond CUSIP and creation calendar date to allow for repeated bond observations and shocks shared by creations on the same day. Tests are two-sided. I report raw returns and returns winsorized at the 0.5th and 99.5th percentiles. For each horizon, I calculate common cutoffs for delivered and control bonds in the pre-match sample with complete covariates and available returns, before replicating controls. Reporting both versions shows how extreme returns affect the estimates without

presuming that those observations are errors.

4.5 ETF prices and NAVs around creation

I trace ETF premiums from ten trading days before to nine trading days after the recorded increase in shares. Each event has equal weight, and each daily mean uses the valid observations available at that offset. For comparison, I follow 24,844 creations with nonnegative initial premiums in the same ETFs and within the date ranges covered by their bond events. These unmatched events provide a descriptive comparison of premium dynamics, not a causal control group.

To compare events with different initial premiums, I scale the premium change by its initial value:

$$G_{e,\tau} = \frac{PD_{e,\tau} - PD_{e,-1}}{PD_{e,-1}}, \quad (5)$$

where τ is the offset in trading days. I require the initial absolute premium to be at least five basis points and winsorize at common 1st and 99th percentile limits for each offset. The measure equals -1 when the premium reaches zero and falls below -1 when the premium changes sign. I also hold event membership fixed in a balanced-window check, since missing observations can change the composition of the daily means.

Finally, I examine prices and NAVs separately. I divide both by each event's NAV on day -1 and multiply by 100. At a given offset, the two means use the same available events. I do not impute missing prices or NAVs. These paths describe valuation changes around creation; they do not isolate the effect of creation on those valuations.

5 Empirical results

5.1 How often do creations follow discounts?

Of the 41,165 net creations with observed initial premiums, 6,445 follow a discount, or 15.66%. This is an event frequency. It gives each creation the same weight, regardless of the number or value of shares issued, and therefore does not measure the fraction of creation volume at a discount. In these events, net primary-market activity runs in the opposite direction to the simple arbitrage prediction based on the initial premium.

The initial discount precedes the recorded share increase; the price when the AP agreed to trade is unobserved. Pre-event premiums alone do not account for these creations. I next examine ETF valuations around the events and the subsequent returns on bonds entering the fund.

5.2 The closing discount

The average discount almost disappears on the creation day. Figure 1 and Table 3 show a mean premium of -13.44 basis points on day -1 , -0.86 on day zero, and 0.65 on day $+1$. By day $+9$, the mean is 0.63 basis points. Most of the adjustment occurs near the recorded creation, with little remaining gap over the rest of the window.

Nonnegative-premium creations show a slower decline in the average premium. It falls from 18.82 basis points before creation to 15.99 on the event day and 14.15 on the next trading day, and remains at 11.38 on day $+9$. The groups differ both in the sign of their initial premium and in the events they contain. Their paths describe different patterns around creation, without identifying the effect of creation relative to redemption or to no primary-market transaction.

Scaling by the initial premium gives a similar result. For discounted creations that meet the five-basis-point threshold, the average scaled change is -0.88 on day zero and

about -1.00 on day $+1$. Thus, the average scaled gap is close to zero by the day after creation. Values below -1 later in the window indicate that the premium has changed sign; they should not be interpreted as progressively larger improvements in pricing efficiency. Positive net issuance accompanies the closing discount, but the paths do not show which party initiates the adjustment.

Both prices and NAVs move toward each other. In Figure 2, baseline NAV is normalized to 100 for each event. The mean ETF price rises from 99.866 before creation to 99.932 on the creation day, while mean NAV falls from 100 to 99.940. The discount therefore closes partly through a lower reported portfolio value. By day $+9$, both series are below their creation-day levels, although the average premium remains close to zero. Convergence to NAV alone says little about the subsequent level of either price or NAV.

5.3 Returns on delivered bonds

Delivered bonds underperform their matched unchanged holdings. Table 4 reports the differences in cumulative clean-price returns at all eight horizons. With winsorization, the difference is -0.16 basis points at horizon 1, -1.36 at horizon 5, and -1.28 at horizon 20. All eight estimates are negative; those at horizons 2 through 20 are significant at the 5% level. Relative prices fall early in the window and remain lower, although the estimated loss does not increase at every horizon.

The regression estimates are similar. At horizon 20, the coefficient on delivery is -1.31 basis points with bond and creation-event fixed effects (Table 5). It becomes -1.36 after adding liquidity controls and remains -1.36 when I also control for prior customer net selling. These controls do not remove the return difference. Selection based on information about future returns that is absent from the data remains possible.

These estimates measure relative bond-price performance. Converting them into an annual return cost for the ETF or a profit for the AP would require the quantities and

exchange valuations of the delivered bonds, the value of the shares issued, acquisition costs, and a counterfactual portfolio. The matched returns also do not establish that delivery caused prices to fall. They show how bonds selected into the portfolio performed relative to similar bonds already held.

5.4 Interpreting the exchange

The ETF discount closes around creation, yet the bonds entering the portfolio subsequently underperform. The first result concerns the pricing of the ETF shares; the second concerns the assets accepted in exchange for those shares. Basket composition remains relevant even after the share price and reported NAV converge.

An AP may choose bonds to reduce inventory, to economize on sourcing costs, or because of private information. The fund’s own cost of acquiring those bonds affects its willingness to accept them. The current estimates do not separate these motives. In particular, controlling for aggregate customer selling does not measure the benefit of reducing the delivering AP’s inventory. Further tests will examine valuation difficulty, the relation between relative bond returns and selling pressure, and the price impact of in-kind creation compared with OTC short sales.

6 Robustness checks

6.1 Sensitivity to large returns

Large returns affect statistical precision more than the sign of the estimates. At horizon 20, the raw paired difference is -0.89 basis points with a standard error of 0.84. Winsorization gives -1.28 with a standard error of 0.63. In the fully controlled regression, the corresponding estimates are -1.10 (standard error 0.68) and -1.36 (standard error 0.49). Neither raw-return estimate is significant at 5% at this horizon. Tables 4 and 5

report both versions.

The raw paired differences are negative at every horizon and significant at horizons 3, 4, and 5. They become less precise later in the window. Evidence of short-run underperformance therefore extends beyond the winsorized horizon-20 estimate, although the tests across horizons and specifications are related and do not include a multiple-testing adjustment. They should be read as a return path, not as independent tests of a particular mechanism.

6.2 Event coverage and matching balance

Changing event coverage does not explain the closing discount. For the 4,416 events with observations at all 20 offsets, the average premium is -13.51 basis points on day -1 , -1.01 on day zero, 0.44 on day $+1$, and 0.27 on day $+9$. The timing and magnitude are close to those obtained using all available observations in Table 3.

Matching also leaves small mean differences in the six bond characteristics. The largest absolute standardized difference is below 0.10 under either pre-match or matched-sample scaling, while event and industry are matched exactly. Balance remains after the 74 pairs with missing or nonpositive liquidity controls are removed. This check concerns the observed characteristics. It cannot establish balance on private information or on the likelihood that the ETF trades a bond directly.

6.3 Holdings and price measurement

Holdings changes can include the fund’s own bond trades. Transfer records would be needed to separate those trades from every in-kind delivery. Infrequent bond trading creates a second concern: some returns rely on stale or forward-filled prices. Winsorization reduces the influence of extreme returns, but leaves stale prices and the exclusion of interest and coupon payments unchanged. These measurement issues limit how closely

the estimated return differences can be tied to the assets actually delivered. The results describe conditional performance around creation, rather than a causal loss imposed on the fund by APs.

7 Conclusion

Corporate bond ETFs can create shares even when their shares have been trading below NAV. Such events account for 15.66% of net creations with observed initial premiums in my sample. The average discount closes rapidly around the recorded creation. Bonds inferred to enter the portfolio then underperform similar holdings whose quantities remain unchanged.

These findings place the choice of bonds alongside the ETF premium in the study of primary-market activity. The discount describes the relative valuation of shares and the reported portfolio; the bond returns describe the performance of assets selected into that portfolio. A closing discount can coexist with selective delivery. AP inventories, the cost of obtaining bonds, and the fund's portfolio needs offer reasons for both parties to accept the exchange.

The evidence is conditional on the observed sample, and longer-horizon precision depends on the treatment of large returns. Daily data leave the terms and motives of transfers unobserved. Further evidence on those decisions is needed to distinguish the benefits of inventory-based intermediation from the cost of informed selection.

References

- Amihud, Y. (2002). Illiquidity and Stock Returns: Cross-Section and Time-Series Effects. *Journal of Financial Markets* 5(1), 31–56.
- Ben-David, I., F. Franzoni, and R. Moussawi (2018). Do ETFs Increase Volatility? *The Journal of Finance* 73(6), 2471–2535.
- Bessembinder, H., S. Jacobsen, W. Maxwell, and K. Venkataraman (2018). Capital Commitment and Illiquidity in Corporate Bonds. *The Journal of Finance* 73(4), 1615–1661.
- Brown, D. C., S. W. Davies, and M. C. Ringgenberg (2021). ETF Arbitrage, Non-Fundamental Demand, and Return Predictability. *Review of Finance* 25(4), 937–972.
- Center for Research in Security Prices (n.d.). CRSP US Stock Databases. Database. Vendor documentation accessed September 26, 2026.
- Dannhauser, C. D. (2017). The Impact of Innovation: Evidence from Corporate Bond Exchange-Traded Funds (ETFs). *Journal of Financial Economics* 125(3), 537–560.
- Dannhauser, C. D. and S. Hoseinzade (2022). The Unintended Consequences of Corporate Bond ETFs: Evidence from the Taper Tantrum. *The Review of Financial Studies* 35(1), 51–90.
- DTCC (2024). ETF Timeline and Submission Processing. Institutional documentation, May 28, 2024; accessed September 26, 2026.
- Du, L. (2025). Inventory in Liquidity Transformation: Evidence from Corporate Bond ETF Creations. Working paper, February 26.
- Duffie, D., N. Gârleanu, and L. H. Pedersen (2005). Over-the-Counter Markets. *Econometrica* 73(6), 1815–1847.
- ETF Global (n.d.). ETF Global: Fund and Constituents Data. Database. WRDS vendor documentation accessed September 26, 2026.

- FINRA (n.d.). TRACE Enhanced Historical Data. Database. Historic Data Information documentation accessed September 26, 2026.
- Goldstein, M. A. and E. S. Hotchkiss (2020). Providing Liquidity in an Illiquid Market: Dealer Behavior in US Corporate Bonds. *Journal of Financial Economics* 135(1), 16–40.
- Hendershott, T., D. Li, D. Livdan, and N. Schürhoff (2020). Relationship Trading in Over-the-Counter Markets. *The Journal of Finance* 75(2), 683–734.
- Huang, A., M. O’Hara, J. Xue, and X. A. Zhou (2026). Beyond Creation and Redemption: Principal Trading by Bond ETFs. Working paper, August; SSRN revision August 4, 2026.
- Koont, N., Y. Ma, Ľ. Pástor, and Y. Zeng (2025). Steering a Ship in Illiquid Waters: Active Management of Passive Funds. *The Review of Financial Studies* 38(10), 2887–2935.
- Lettau, M. and A. Madhavan (2018). Exchange-Traded Funds 101 for Economists. *Journal of Economic Perspectives* 32(1), 135–154.
- Pan, K. and Y. Zeng (2023). ETF Arbitrage under Liquidity Mismatch. Working paper, August.
- Reilly, C. (2022). The Hidden Cost of Corporate Bond ETFs. Working paper, June 27.
- Shim, J. J. and K. Todorov (2023). ETFs, Illiquid Assets, and Fire Sales. Working paper, October 10; SSRN revision November 27, 2023.

Table 1: Sample construction

<i>Panel A. ETF panel and creation frequency</i>				Count
ETF-date observations				255,044
ETFs				211
Net creations				41,267
Net creations with observed initial premiums				41,165
Net creations following discounts				6,445
Share following discounts				15.66%
<i>Panel B. Bond sample construction</i>				
	Delivered	Controls	Events	
Eligible bond-events	889,882	6,568,638	4,886	
Complete matching covariates	623,390	5,211,366	4,880	
Initial matched pairs	599,235	599,235	4,319	
Complete matched returns	503,224	503,224	4,292	
Additional liquidity controls	503,150	503,150	4,290	

The ETF panel covers January 4, 2016, through October 30, 2025. A creation is a positive net change in shares outstanding; its initial premium is measured on the preceding trading day. Panel A omits 102 creations with missing premiums when calculating the frequency. Panel B begins with eligible bond-events drawn from 4,902 source creation events. Each delivered bond is matched to an unchanged holding in the same ETF event and industry, with replacement. The last three rows count reused controls at their matching frequencies. The final two rows require returns at all eight horizons, and the last row also requires the regression liquidity controls. The bond sample covers fewer events than the ETF frequency sample.

Table 2: Bond characteristics after matching

Characteristic	Delivered	Control	SMD
Log offering amount	13.815	13.767	0.076
Remaining maturity (years)	9.484	9.197	0.031
Coupon (%)	4.205	4.178	0.017
Average numerical rating	8.339	8.359	-0.008
Log(1 + turnover)	4.162	4.103	0.055
Customer net selling (USD millions)	-0.082	-0.069	-0.002

The table describes 503,224 pairs of delivered and control bonds with finite returns at all eight horizons. Controls are selected with replacement within the same ETF creation event and industry using the six pre-event characteristics listed. Reused controls enter the means with their matching frequencies. SMD is the difference between delivered and control means divided by the fixed pooled standard deviation before matching. Customer net selling equals customer sales to dealers less customer purchases, averaged over the preceding ten days. Higher numerical ratings indicate worse credit quality. The largest absolute SMD is 0.076 with pre-match scaling and 0.081 with matched-sample scaling. Industry frequencies are identical because industry is matched exactly.

Table 3: ETF premiums around creation

Event day	Discounted creations		Nonnegative-premium creations	
	Premium (bp)	Events	Premium (bp)	Events
-1	-13.44	4,868	18.82	24,844
0	-0.86	4,869	15.99	24,844
+1	0.65	4,833	14.15	24,707
+9	0.63	4,795	11.38	24,629
<i>Balanced-window means: discounted creations</i>				
-1	-13.51	4,416		
0	-1.01	4,416		
+1	0.44	4,416		
+9	0.27	4,416		

The premium is $10,000(P/NAV - 1)$. Day zero records the increase in shares outstanding; offsets follow the project trading calendar. Daily means give equal weight to all events with valid observations at that offset. The discounted group includes the source bond events, of which 34 lack cleaned baseline premiums but may have observations at other offsets. Nonnegative-premium creations come from the same ETFs within their source-event date ranges. The balanced sample requires a valid premium at every offset from -10 through $+9$. The groups are not matched counterfactuals.

Table 4: Matched bond returns

Horizon	Raw returns		Winsorized returns	
	Difference	Std. error	Difference	Std. error
1	-0.197	0.230	-0.156	0.183
2	-0.399	0.280	-0.453**	0.200
3	-1.177***	0.309	-1.060***	0.230
4	-1.149**	0.456	-1.126***	0.291
5	-1.230**	0.542	-1.364***	0.327
10	-1.000	0.657	-1.343***	0.422
15	-0.939	0.761	-1.179**	0.589
20	-0.892	0.837	-1.281**	0.632

The table reports mean differences between delivered and control cumulative clean-price returns, in basis points, measured from the recorded creation date. Every column uses the same 503,224 pairs. Horizons follow the source bond-price series, which uses last-trade prices and forward fills; they have not been reconstructed as counts of trading days on a common calendar. At each horizon, winsorization applies common 0.5th and 99.5th percentile cutoffs to both groups, calculated before controls are replicated. Standard errors are clustered by CUSIP and creation date. Stars denote two-sided Student t tests with 1,427 degrees of freedom: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. No adjustment is made for testing across horizons.

Table 5: Conditional bond performance at horizon 20

	(1)	(2)	(3)
<i>Raw returns</i>			
Inferred delivery	-1.025 (0.685)	-1.096 (0.680)	-1.095 (0.680)
<i>Winsorized returns</i>			
Inferred delivery	-1.309*** (0.488)	-1.360*** (0.489)	-1.359*** (0.489)
Bond and event fixed effects	Yes	Yes	Yes
Liquidity controls	No	Yes	Yes
Prior customer net selling	No	No	Yes
Pairs	503,224	503,150	503,150
Observations	1,006,448	1,006,300	1,006,300
Creation events	4,292	4,290	4,290
R^2 , raw	0.633	0.633	0.633
R^2 , winsorized	0.698	0.699	0.699

The table reports the coefficient on inferred delivery in equation (4). The dependent variable is the cumulative clean-price return in basis points. Each matched pair contributes two observations, and controls are repeated when matched more than once. All specifications include bond and ETF creation-event fixed effects. Liquidity controls are pre-event log turnover and log Amihud illiquidity. Column (3) also includes customer net selling over the preceding ten days. Columns (2) and (3) lose 74 pairs because Amihud illiquidity is missing or nonpositive. Parentheses contain standard errors clustered by CUSIP and creation date. Stars denote two-sided Student t tests with 1,427 degrees of freedom in column (1) and 1,426 in columns (2) and (3): *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Within each column, raw and winsorized estimates use the same sample.

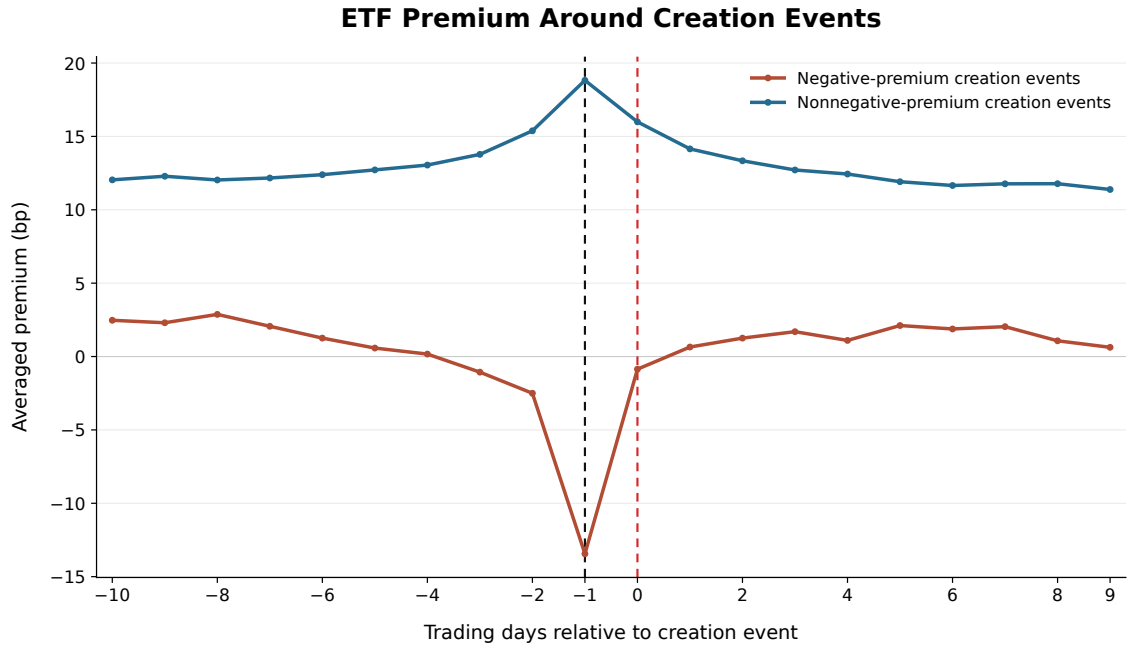
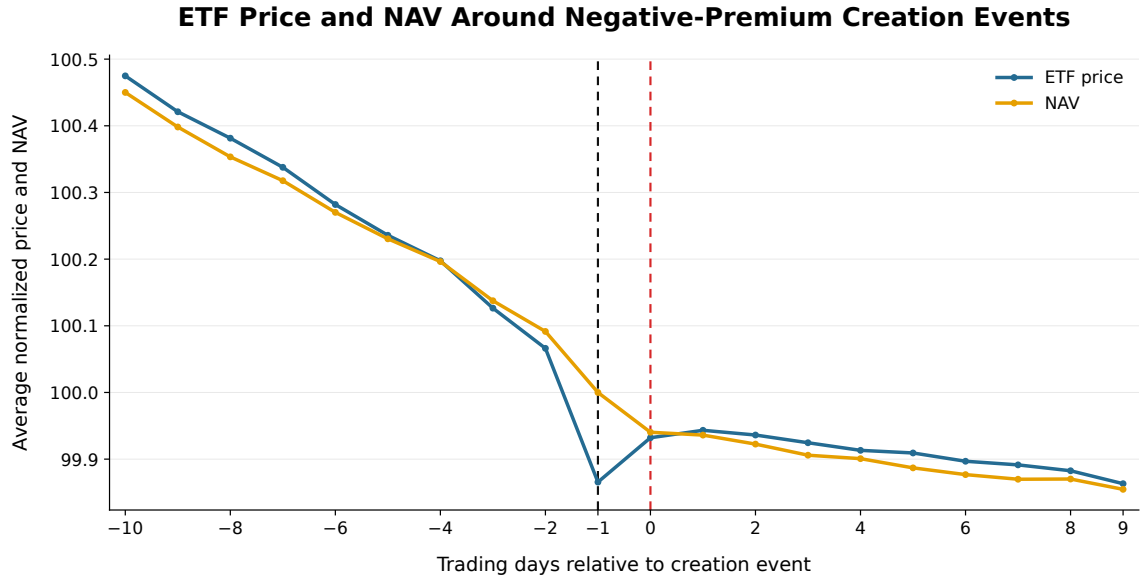


Figure 1: ETF premiums around creation

The figure follows mean ETF premiums, in basis points, from ten trading days before to nine trading days after the recorded share increase (day zero). Each event has equal weight. The discounted group consists of the source bond events; nonnegative-premium creations come from the same ETFs within their bond-event date ranges. Daily means use all valid observations available at each offset, allowing event counts to vary. The groups are observational and are not randomly assigned. Table 3 reports selected means and event counts.



$$P_{i,t}^* = 100 \frac{P_{i,t}}{NAV_{i,-1}}$$

$$NAV_{i,t}^* = 100 \frac{NAV_{i,t}}{NAV_{i,-1}}$$

Figure 2: ETF prices and NAVs around discounted creation
Prices and NAVs are divided by each event's NAV on the trading day before creation and multiplied by 100, placing both series on the same scale within an event. At each offset, the means use the same available events with equal weights. No missing prices or NAVs are imputed. The 4,902 source events yield between 4,792 and 4,869 valid observations per offset. The ratio of mean price to mean NAV can differ from the mean event premium. Both series measure levels; neither is a total-return series.

A Sample construction and variable definitions

A.1 Sample restrictions and matching

Table 1 distinguishes the sample used to measure creation frequency from the sample used to estimate bond returns. Screening removes 10,327 ETF records and leaves 255,044 observations. Premiums are available for 254,601. Of the 41,267 net creations, 102 have missing premiums, leaving 41,165 in the frequency denominator. The 6,445 discounted creations have strictly negative pre-event premiums; zero premiums belong to the non-negative group.

The bond analysis begins with 4,902 creation events. Identifier and eligibility checks leave 4,886 in the eligible bond panel. Matching requires a reported industry and finite values for all six covariates. Among 623,390 eligible delivered bond-events with complete covariates, 24,155 have no eligible unchanged holding in the same industry. The remaining 599,235 form matched pairs. Missing returns remove another 96,011 pairs. I keep the original matches when applying the return requirement, without searching again for controls. The final sample contains 14,832 distinct delivered bonds, 14,715 distinct control bonds, and 360,643 distinct control bond-events. A bond can belong to both groups across different events.

For characteristic k , the matching scale is the square root of the average delivered and control variances in the complete-covariate sample before matching. When controls have the same distance, CUSIP order breaks the tie. A control can be selected repeatedly; both the balance calculations and regressions count it with its matching frequency. Matching uses neither propensity scores nor a distance cutoff for the creation sample.

A.2 Bond characteristics and inference

Offering amount enters in natural logs. Remaining maturity is measured in years and coupon in percentage points. The rating is the average of the numerical S&P and Moody’s ratings, so larger values indicate worse credit quality. Turnover is dollar trading volume over the preceding 30-day window divided by offering amount, transformed as $\log(1 + \text{turnover})$. Customer net selling is customer sales to dealers less customer purchases from dealers, measured in dollars and averaged over the preceding ten days. It is positive when customers sell on net and measures a flow rather than a dealer inventory position.

Amihud illiquidity enters the regressions but not the match. Requiring an observed positive value removes both bonds in 74 pairs. The specifications with this control use 503,150 pairs, or 1,006,300 observations after counting reused controls at their matching frequencies. They contain 15,530 bonds and 4,290 creation events. The fixed-effects-only specification uses 503,224 pairs and 4,292 events. Inference is two-sided, with standard errors clustered by bond and creation date. Student t tests use the smaller cluster count minus one as the degrees of freedom. These standard errors condition on the chosen matches and do not account for uncertainty in matching.

A.3 Event-time coverage

The discounted-creation paths retain the 4,902 source events, including 34 with missing cleaned baseline premiums. An event can lack a baseline premium but have a valid observation at another offset, so the number of events may change along the path. The balanced check requires observations at every offset from -10 through $+9$. For scaled premium changes, requiring an initial absolute premium of at least five basis points leaves 2,894 discounted creations and 19,948 nonnegative-premium creations before checking endpoint availability. Winsorization cutoffs are calculated at each offset using the valid ETF-date reference panel and applied to both groups.